Appendices

APPENDIX A

Cost estimation of the project:

Items	**Cost (Taka)**
Software:	
i) Python 3.11 (programming environment)	Free/Open Source
ii) PyTorch and DeepXDE (modeling and PINN libraries)	Free/Open Source
iii) NumPy, pandas, SciPy, scikit-learn, matplotlib, seaborn	Free/Open Source
iv) `uv` (dependency management)	Free/Open Source
v) LaTeX/MiKTeX (typesetting)	Free/Open Source
Databases and Data Sources:	
i) IMS Bearing Dataset (NASA Prognostics Data Repository)	Free/Publicly available
ii) Reference papers and literature (secondary data)	Free/Publicly available
Hardware/Computing Resources:	
i) Laptop (AMD Ryzen 7 5800H, NVIDIA RTX 3060 Laptop GPU) used for model training and analysis	Already Available
ii) Local storage for datasets, features, and figures	Already Available
Miscellaneous:	
i) Documentation and report writing tools (LaTeX)	Free/Open Source
ii) Version control (Git/GitHub)	Free
iii) Reference management (Zotero/Mendeley)	Free
Total Estimated Cost:	**Negligible**

APPENDIX B

Gantt chart of the project:

APPENDIX C

The originality and AI reports generated by Turnitin are attached below.

MAHDEE HASSAN

ORIGINALITY REPORT

6%	3%	4%	3%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Chittagong University of Engineering and Technology Student Paper	<1%
2	Mohammed Chalouli, Nasr-eddine Berrached, Mouloud Denai. "Intelligent Health Monitoring of Machine Bearings Based on Feature Extraction", Journal of Failure Analysis and Prevention, 2017 Publication	<1%
3	arxiv.org Internet Source	<1%
4	Jacek Dybała. "Diagnosing of rolling-element bearings using amplitude level-based decomposition of machine vibration signal", Measurement, 2018 Publication	<1%
5	coek.info Internet Source	<1%
6	Submitted to Liverpool John Moores University Student Paper	<1%
7	ojs.istp-press.com Internet Source	<1%

0% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

0 AI-generated only 0%
Likely AI-generated text from a large-language model.

0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

APPENDIX D

CO-PO-K-P-A Mapping

(This part will be filled/checked by the supervisor)

PO1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Title of the Project: Physics-Informed Neural Networks for Remaining-Useful-Life Prediction of Rolling-Element Bearings

Student ID: 2003101

		CO-PO-K-P-A Mapping																															
		Program Outcomes												**Knowledge Profile**								**Complex Engineering Problem Solving**							**Complex Engineering Activities**				
		PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**K1**	**K2**	**K3**	**K4**	**K5**	**K6**	**K7**	**K8**	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**A1**	**A2**	**A3**	**A4**	**A5**
		Needs complex engineering problem solution (P1 + another P)																											Related to project				
Course No.	Course Title	Engineering Knowledge	Problem Analysis	Design/Development of Solutions	Investigation	Modern Tool Usage	The Engineers & Society	Environment and Sustainability	Ethics	Individual and team work	Communication	Project Management and Finance	Life-long Learning	Science	Math	Engineering fundamentals	Engineering specialization	Design	Technology	Society	Research	Knowledge K3,K6,K8	Wide ranging/conflicting	No obvious solution	Infrequent issues	Outside problems	Diverse groups	Many components	Range of resources	Level of interaction	Innovation	Consequences	Familiarity
ME 498	Project and Thesis																																